

```
In [1]: # Preprocesado y modelo
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import os
```

```
In [15]: # Cargar la base de datos
dataF = pd.read_excel("Amazon.xlsx")
dataF.head()
```

```
Out[15]:
```

	Unnamed: 0	Velocidad Entrega	Precio	Durabilidad	Imagen Producto	Valor Educativo	Servicio Retorno	Tamano Paquete	Ci Pro
0	Adam	205	3	345	235	24	23	26	
1	Anna	9	15	315	33	25	4	42	
2	Bernard	17	26	285	3	43	27	41	
3	Edward	135	5	355	295	18	23	39	
4	Emilia	3	45	48	39	34	46	225	

```
In [19]: # 1. Separar las columnas con valores numéricos
data = dataF.select_dtypes(include=['number'])
data
```

```
Out[19]:
```

	Velocidad Entrega	Precio	Durabilidad	Imagen Producto	Valor Educativo	Servicio Retorno	Tamano Paquete	Calidad Producto	Nu Est
0	205	3	345	235	24	23	26	21	
1	9	15	315	33	25	4	42	215	
2	17	26	285	3	43	27	41	26	
3	135	5	355	295	18	23	39	195	
4	3	45	48	39	34	46	225	34	
...	...	...	...	...	...	...	...	...	
95	3	8	32	25	7	21	42	17	
96	305	25	46	24	33	28	355	26	
97	1	14	26	25	24	27	42	185	
98	155	11	335	34	26	29	42	215	
99	125	9	45	25	22	3	3	22	

100 rows × 9 columns

```
In [21]: # 2. Separar la(s) columna(s) con texto o nombres
data_nombres = dataF.select_dtypes(include=['object', 'string'])
data_nombres
```

```
Out[21]:
```

Unnamed: 0	
0	Adam
1	Anna
2	Bernard
3	Edward
4	Emilia
...	...
95	Teofan
96	Teofil
97	Teofila
98	Teon
99	Teresa

100 rows × 1 columns

```
In [23]: from sklearn.preprocessing import normalize
data_scaled = normalize(data)
data_scaled
```

```
Out[23]: array([[0.43826336, 0.00641361, 0.73756517, 0.50239946, 0.05130888,
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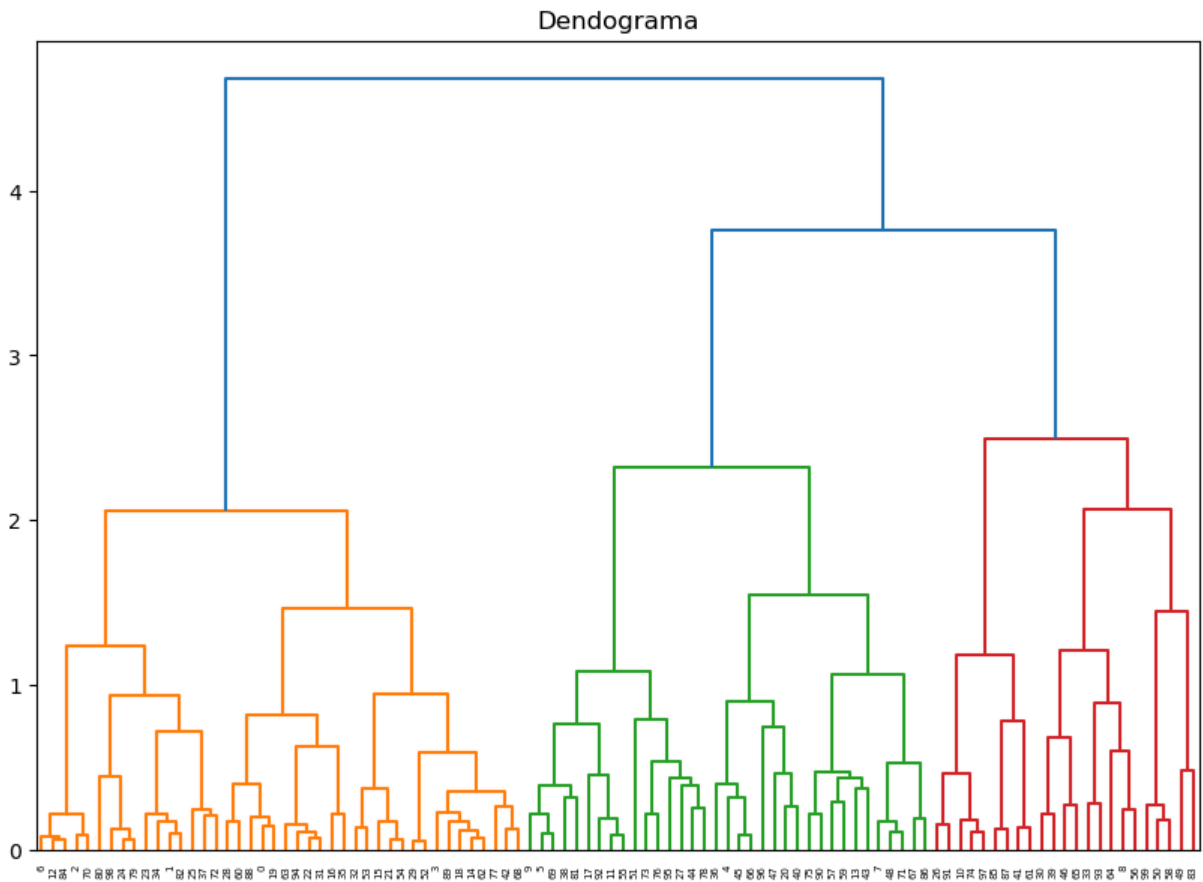
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```

```
In [25]: data_scaled = pd.DataFrame(data_scaled, columns = data.columns)
data_scaled.head()
```

	Velocidad Entrega	Precio	Durabilidad	Imagen Producto	Valor Educativo	Servicio Retorno	Tamano Paquete	Calidad Producto	N E
0	0.438263	0.006414	0.737565	0.502399	0.051309	0.049171	0.055585	0.044895	0.
1	0.023235	0.038725	0.813234	0.085196	0.064542	0.010327	0.108431	0.555065	0.
2	0.057235	0.087535	0.959520	0.010100	0.144770	0.090902	0.138036	0.087535	0.
3	0.258856	0.009587	0.680696	0.565649	0.034514	0.044101	0.074781	0.373904	0.
4	0.011975	0.179625	0.191600	0.155675	0.135717	0.183617	0.898127	0.135717	0.

```
In [28]: import scipy.cluster.hierarchy as shc
plt.figure(figsize=(10,7))
plt.title("Dendograma")
dend = shc.dendrogram(shc.linkage(data_scaled, method='ward'))
```



```
In [30]: print(dend['color_list'])
```

```
['C1', 'C1', 'C1', 'C1', 'C1', 'C1', 'C1', 'C1', 'C1', 'C1', 'C1', 'C1', 'C1', 'C1', 'C1',
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'C0']
```

```
In [32]: colores_unicos = set(dend['color_list'])
colores_unicos
```

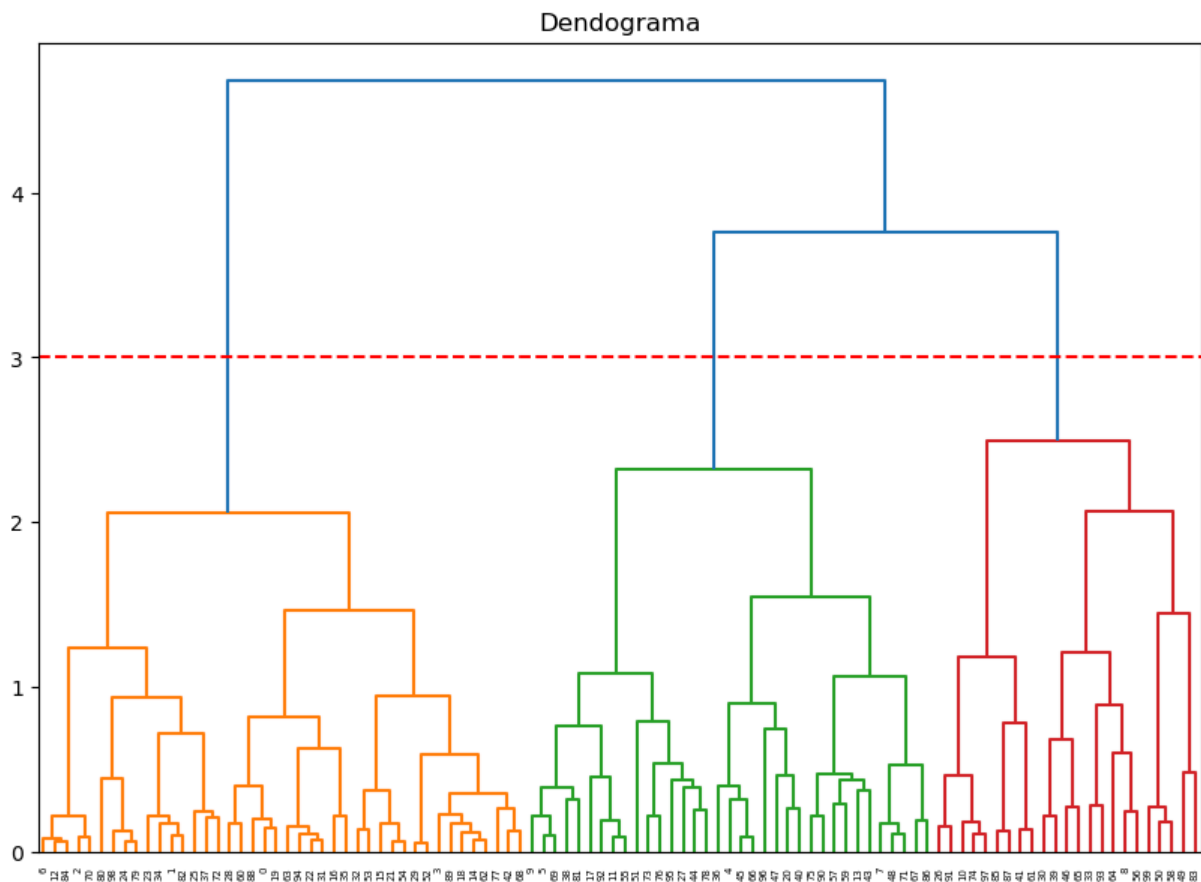
```
Out[32]: {'C0', 'C1', 'C2', 'C3'}
```

```
In [34]: num_clusters_optimo = len(colores_unicos) - 1
num_clusters_optimo
```

```
Out[34]: 3
```

```
In [38]: plt.figure(figsize=(10,7))
plt.title("Dendrogram")
dend = shc.dendrogram(shc.linkage(data_scaled, method='ward'))
plt.axhline(y=3, color = 'r', linestyle = '--')
```

```
Out[38]: <matplotlib.lines.Line2D at 0x1d199ca1850>
```

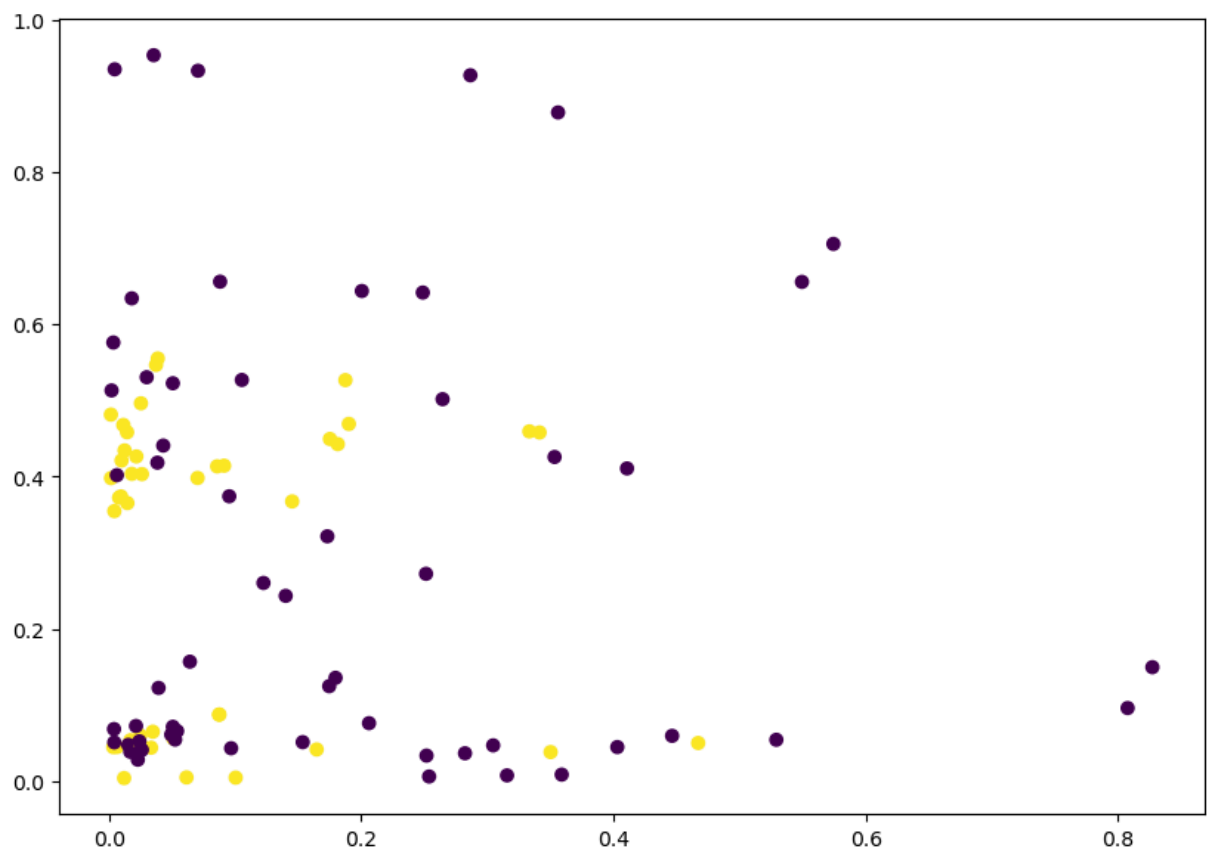


```
In [40]: from sklearn.cluster import AgglomerativeClustering
cluster = AgglomerativeClustering(n_clusters=2, metric='euclidean', linkage='ward')
grupos = cluster.fit_predict(data_scaled)
grupos
```

```
Out[40]: array([1, 1, 1, 1, 0, 0, 1, 0, 0, 0, 0, 0, 1, 0, 1, 1, 1, 0, 1, 1, 0, 1,
        1, 1, 1, 1, 0, 0, 1, 1, 0, 1, 1, 0, 1, 1, 0, 1, 0, 0, 0, 0, 1, 0,
        0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 1, 1, 0, 0, 0, 0, 0, 1, 0, 1, 1, 0, 0,
        0, 0, 1, 0, 1, 0, 1, 0, 0, 0, 0, 1, 0, 1, 1, 0, 1, 0, 1, 0, 0, 0,
        1, 1, 0, 0, 0, 0, 1, 0, 0, 0, 1, 0], dtype=int64)
```

```
In [48]: plt.figure(figsize = (10,7))
plt.scatter(data_scaled['Precio'], data_scaled['Calidad Producto'], c = cluster.lab
```

```
Out[48]: <matplotlib.collections.PathCollection at 0x1d19e62bd10>
```



```
In [50]: # Separación de grupos mediante PCA
campos = data_scaled.values
print(campos)
```

[[0.43826336 0.00641361 0.73756517 0.50239946 0.05130888 0.04917101
0.05558462 0.04489527 0.03634379]
[0.02323527 0.03872544 0.81323434 0.08519598 0.06454241 0.01032679
0.10843125 0.55506471 0.0722875]
[0.05723452 0.08753514 0.95951982 0.01010021 0.14476966 0.09090188
0.13803618 0.08753514 0.11110229]
[0.25885639 0.00958727 0.68069644 0.56564916 0.03451419 0.04410146
0.07478074 0.37390368 0.03259673]
[0.01197502 0.17962533 0.19160036 0.15567529 0.13571692 0.18361701
0.89812667 0.13571692 0.17164198]
[0.14494006 0.251738 0.60264551 0.03661644 0.03966781 0.02898801
0.73995715 0.03356507 0.00457705]
[0.04755854 0.02481315 0.98218723 0.06823617 0.07237169 0.09304932
0.07857498 0.05996511 0.06410064]
[0.13323393 0.04304481 0.06354234 0.52268696 0.05739308 0.04509456
0.70716471 0.44069685 0.05944283]
[0.73804829 0.0214705 0.12613916 0.63069581 0.09393342 0.00805144
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0.75708755 0.04699164 0.06787681]
[0.05320841 0.03547227 0.19509749 0.10641681 0.00886807 0.12415295
0.12858698 0.9533173 0.07537858]
[0.30081705 0.01696917 0.70190646 0.03548099 0.00462795 0.03856629
0.6402004 0.03856629 0.04936485]
[0.03421592 0.01710796 0.98981777 0.0464359 0.05132388 0.03421592
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[0.02878814 0.00169342 0.8213087 0.3979537 0.04572234 0.02878814
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0.07062951 0.05012417 0.05240254]
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0.04853005 0.42106953 0.05566682]
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0.05255293 0.00463702 0.0061827]
[0.386348 0.10536764 0.10068463 0.00468301 0.04917156 0.04214705
0.73757346 0.52683818 0.06087908]
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[0.05696103 0.35600642 0.15664282 0.11392205 0.09018829 0.11866881
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[0.03874633 0.09686582 0.72276808 0.45452426 0.04768779 0.05811949
0.49923156 0.04321706 0.05811949]

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[0.00737441 0.14011373 0.52358287 0.3318483 0.02212322 0.04572132
0.73006626 0.24335542 0.03539715]
[0.40430304 0.01437522 0.78165255 0.04132876 0.05570397 0.03773495
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[0.44633507 0.01460733 0.70602093 0.0308377 0.05842932 0.03408377
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[0.50194895 0.8074831 0.16586139 0.10911934 0.01309432 0.10911934
0.16149662 0.09602502 0.09602502]

```
[0.03014465 0.03478229 0.98549814 0.00695646 0.06492694 0.06492694
 0.07883985 0.06492694 0.08811513]
[0.44300067 0.54932083 0.12404019 0.07442411 0.09923215 0.07796812
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 0.11813124 0.70550604 0.06890989]
[0.29097875 0.0120405 0.73246376 0.61205876 0.0040135 0.05016875
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 0.09775176 0.92695635 0.08089801]
[0.36452026 0.04938662 0.59969463 0.091718 0.08466277 0.00940697
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 0.06149192 0.00491935 0.03935483]
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[0.89135765 0.06417775 0.32088875 0.17827153 0.15687895 0.02139258
 0.02139258 0.15687895 0.1283555 ]]
```

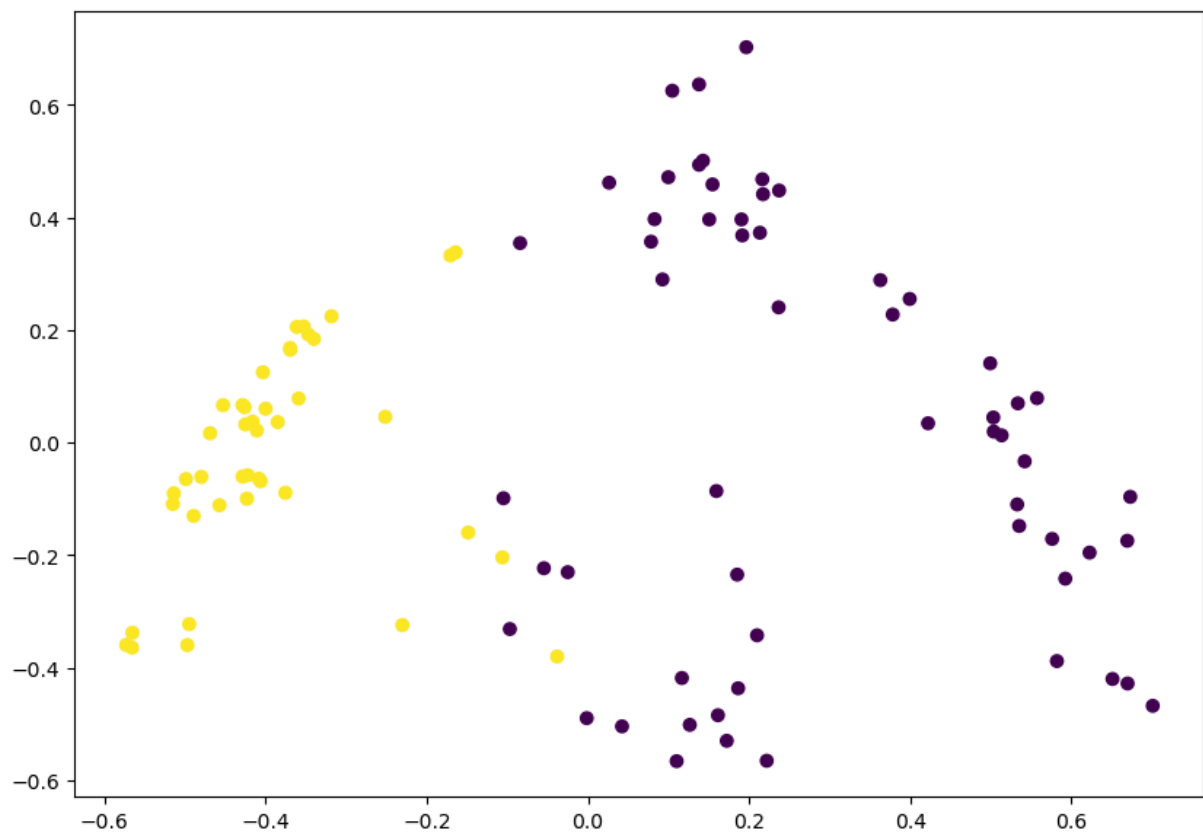
```
In [52]: from sklearn import decomposition
pca = decomposition.PCA(n_components = 2)
pca.fit(campos)
campos = pca.transform(campos)
campos
```

```
Out[52]: array([[ -0.39993219,  0.06035374],
 [ -0.40829583, -0.06402268],
 [ -0.49703182, -0.35955286],
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 [  0.04236318, -0.50393996],
```

```
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[ 0.19164058,  0.36824856],
[-0.38487326,  0.03658414],
[-0.08413049,  0.3546116 ]])
```

```
In [54]: plt.figure(figsize=(10,7))
plt.scatter(campos[:,0], campos[:,1], c = cluster.labels_)
```

```
Out[54]: <matplotlib.collections.PathCollection at 0x1d19e4ecf90>
```



```
In [56]: dataframe = pd.DataFrame(grupos, columns=['grupo'])
dataframe
```

```
Out[56]:
```

	grupo
0	1
1	1
2	1
3	1
4	0
...	...
95	0
96	0
97	0
98	1
99	0

100 rows × 1 columns

```
In [58]: dataframe2 = pd.concat([data, dataframe], axis=1, join = 'inner')
dataframe2
```

Out[58]:

	Velocidad Entrega	Precio	Durabilidad	Imagen Producto	Valor Educativo	Servicio Retorno	Tamano Paquete	Calidad Producto	Nu Esti
0	205	3	345	235	24	23	26	21	
1	9	15	315	33	25	4	42	215	
2	17	26	285	3	43	27	41	26	
3	135	5	355	295	18	23	39	195	
4	3	45	48	39	34	46	225	34	
...	...	...	...	...	...	...	...	...	
95	3	8	32	25	7	21	42	17	
96	305	25	46	24	33	28	355	26	
97	1	14	26	25	24	27	42	185	
98	155	11	335	34	26	29	42	215	
99	125	9	45	25	22	3	3	22	

100 rows × 10 columns



In [62]:

```
dataframeFinal= pd.concat([data_nombres, dataframe2], axis=1, join = 'inner')
dataframeFinal
```

Out[62]:

	Unnamed: 0	Velocidad Entrega	Precio	Durabilidad	Imagen Producto	Valor Educativo	Servicio Retorno	Tamano Paquete	Pr
0	Adam	205	3	345	235	24	23	26	
1	Anna	9	15	315	33	25	4	42	
2	Bernard	17	26	285	3	43	27	41	
3	Edward	135	5	355	295	18	23	39	
4	Emilia	3	45	48	39	34	46	225	
...	...	...	...	...	...	...	...	...	
95	Teofan	3	8	32	25	7	21	42	
96	Teofil	305	25	46	24	33	28	355	
97	Teofila	1	14	26	25	24	27	42	
98	Teon	155	11	335	34	26	29	42	
99	Teresa	125	9	45	25	22	3	3	

100 rows × 11 columns



```
In [64]: pd.options.display.max_rows = None  
dataframeFinal
```

Out[64]:

	Unnamed: 0	Velocidad Entrega	Precio	Durabilidad	Imagen Producto	Valor Educativo	Servicio Retorno	Tamano Paquete	Pi
0	Adam	205	3	345	235	24	23	26	
1	Anna	9	15	315	33	25	4	42	
2	Bernard	17	26	285	3	43	27	41	
3	Edward	135	5	355	295	18	23	39	
4	Emilia	3	45	48	39	34	46	225	
5	Fabian	95	165	395	24	26	19	485	
6	Philip	23	12	475	33	35	45	38	
7	Frank	65	21	31	255	28	22	345	
8	Xavier	275	8	47	235	35	3	38	
9	Gabriel	2	175	325	3	37	32	435	
10	Marisol	12	8	44	24	2	28	29	
11	Henry	195	11	455	23	3	25	415	
12	Irene	14	7	405	19	21	14	33	
13	Isabelle	185	75	43	285	27	37	335	
14	Isidore	235	65	495	335	3	26	34	
15	Joseph	17	1	485	235	27	17	24	
16	Eugene	16	205	285	255	36	29	31	
17	Eugenia	245	9	385	215	34	15	295	
18	Eunice	265	7	485	305	33	39	34	
19	Eva	235	65	495	335	3	26	34	
20	Evdokia	165	45	43	2	21	18	315	
21	Fedir	17	2	415	125	12	17	26	
22	Felix	15	2	455	355	35	34	42	
23	Fialka	12	75	335	24	19	25	36	
24	Flavia	255	7	435	24	33	26	19	
25	Flora	23	105	395	29	34	28	235	
26	Florent	12	75	33	24	19	25	36	
27	Florence	26	65	485	305	32	39	335	
28	Hannah	175	14	495	175	31	17	27	
29	Helen	205	185	295	275	39	3	42	

	Unnamed: 0	Velocidad Entrega	Precio	Durabilidad	Imagen Producto	Valor Educativo	Servicio Retorno	Tamano Paquete	Pi
30	Herman	15	16	3	265	31	3	4	
31	Hilary	14	19	445	345	33	32	41	
32	Lourdes	26	1	465	295	37	24	23	
33	Isadore	17	185	32	285	35	34	42	
34	Ivan	12	5	385	17	17	11	31	
35	Jacob	9	165	375	225	25	24	38	
36	Jeremiah	18	2	29	29	37	25	465	
37	Jervis	2	45	455	27	24	26	365	
38	Joachim	0	105	345	27	11	26	445	
39	John	12	1	32	225	21	22	44	
40	Santiago	95	17	38	23	26	25	385	
41	Josephine	295	45	48	39	34	46	225	
42	Judith	245	115	465	225	36	13	31	
43	Justin	25	65	43	235	31	25	185	
44	Kalyna	1	13	325	185	24	17	425	
45	Larissa	25	125	47	23	37	14	315	
46	Lawrence	155	95	5	225	26	32	19	
47	Leon	17	195	28	28	36	23	455	
48	Leonard	29	1	44	225	3	24	335	
49	Leonid	27	105	4	15	38	14	26	
50	Lesia	185	35	41	3	21	25	26	
51	Leo	13	24	41	25	36	25	45	
52	Louise	225	205	315	295	43	34	44	
53	Lubomyr	14	12	335	245	25	26	46	
54	Lydia	19	4	435	145	16	21	28	
55	Magdalyna	145	13	385	35	28	36	385	
56	Maksym	245	22	37	345	46	4	48	
57	Marcel	27	125	48	275	4	3	385	
58	Margaret	215	9	38	27	31	25	22	
59	Maria	115	225	4	235	33	22	435	

	Unnamed: 0	Velocidad Entrega	Precio	Durabilidad	Imagen Producto	Valor Educativo	Servicio Retorno	Tamano Paquete	Pi
60	Marian	155	95	495	225	26	31	19	
61	Marianna	255	95	46	29	36	23	225	
62	Markian	205	55	465	275	25	27	37	
63	Marko	15	19	275	245	34	26	3	
64	Martha	55	1	36	235	16	32	5	
65	Martin	185	7	45	225	26	23	34	
66	Maryna	21	125	46	31	33	39	365	
67	Matthew	8	225	32	265	3	25	355	
68	Maura	265	85	425	185	35	19	24	
69	Maya	115	185	415	26	3	23	455	
70	Maximillian	18	27	295	31	45	29	42	
71	Melania	28	11	41	155	4	16	265	
72	Methodius	18	11	495	24	29	19	245	
73	Michael	26	65	455	225	33	27	365	
74	Michaelina	15	1	33	33	24	27	41	
75	Mina	21	12	47	245	32	27	425	
76	Monica	19	4	415	305	22	26	265	
77	Mykyta	165	13	485	165	29	15	26	
78	Myron	5	95	355	225	15	31	495	
79	Myroslav	225	8	435	23	31	21	34	
80	Myroslava	275	9	435	19	36	21	245	
81	Salome	17	23	275	41	4	44	315	
82	Samuel	8	14	305	32	23	38	41	
83	Sandra	115	185	38	25	3	25	37	
84	Sarah	13	15	425	3	28	28	34	
85	Savina	125	155	35	21	28	22	45	
86	Sebastian	12	145	42	295	27	27	335	
87	Sophia	105	175	37	24	28	23	36	
88	Stephan	145	6	365	305	2	25	4	
89	Stephania	215	125	465	315	34	4	37	

	Unnamed: 0	Velocidad Entrega	Precio	Durabilidad	Imagen Producto	Valor Educativo	Servicio Retorno	Tamano Paquete	Pi
90	Susanna	15	14	39	355	3	38	395	
91	Sylvan	24	85	38	21	33	14	29	
92	Sylvester	155	21	255	39	36	4	295	
93	Tamara	95	135	25	245	22	25	41	
94	Theodore	2	25	335	225	22	21	25	
95	Teofan	3	8	32	25	7	21	42	
96	Teofil	305	25	46	24	33	28	355	
97	Teofila	1	14	26	25	24	27	42	
98	Teon	155	11	335	34	26	29	42	
99	Teresa	125	9	45	25	22	3	3	

Salomé comparte un perfil de evaluación similar con los integrantes del Grupo 0. Este grupo se caracteriza por dar menor peso o puntuación a la Durabilidad y priorizar variables como el Tamaño del Paquete o la Velocidad de Entrega.

Stephanía forma parte del Grupo 1, cuyos miembros muestran preferencias de evaluación alineadas. Este clúster se distingue por otorgar calificaciones significativamente altas a la Durabilidad del producto

Al estar ubicada en el Grupo 1, sus patrones de valoración coinciden con los usuarios que priorizan la resistencia y calidad duradera de los artículos.